06D — Deep-Dive Feature Audit: Pills & Alchemy (Part 4 of 6)

Pre-Fight Combat Buff Pills (`P:CB_*`), Tactical/Stealth/Control-Resilience Pills, Beast-Bond Pills & Forbidden Blood-Sacrifice Overdrive Pills

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Pre-battle elixir preparation, scout stealth pills, spirit-beast stamina recovery, and desperate forbidden lifespan-burning pills must arise organically from combat stakes, sect orthodoxy laws, physical logistics (`!away`), and NPC personality—never wasted on friendly spars or blocked by eating a bowl of rice!
- **Immutable Project Rules**:
    1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
    2. **Preserve exact 40 `<script>` tags** and load order in `index.html` / `test/load.js`.
    3. **Strict Closed-Loop Resource Conservation** & **Zero RNG consumption in read/score/UI paths**.

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1. Architectural Scope of Group 4 (`06D`)

Group 4 audits all combat-enhancing, tactical-preparation, pet-synergy, and taboo overdrive pills:

1. **Six Canonical Pre-Fight Combat Pills (`data/data_cb.js` — `P:CB_*`)**:
    - `CB_FURY` (โอสถเดือดโลหิต, Tier 2: `+25% atk, +15% matk, +6 crit`, `tox: 5`)
    - `CB_STONE` (โอสถผิวศิลา, Tier 2: `+30% def, +12% mdef, +10% hp`, `tox: 4`)
    - `CB_WIND` (โอสถย่างลม, Tier 2: `+25% eva, +20 move, +12 cast`, `tox: 4`)
    - `CB_EYE` (โอสถเนตรแจ่ม, Tier 3: `+30% acc, +8 crit, +10 pen`, `tox: 6`)
    - `CB_SPIRIT` (โอสถห่มจิตสวรรค์, Tier 3: `+25% matk, +25% mdef, +8 cast`, `tox: 6`)
    - `CB_BERSERK` (โอสถคลั่งสวรรค์, Tier 4: `+45% atk, +25% matk, -20% def, +12 crit, +30 critDmg`, `tox: 14`)

2. **Specialized Tactical, Beast & Taboo Pills (`data/data_items.js` & `v2.0.34+`)**:
    - `ITEM190_PILL_009` (โอสถใจกระจ่างต้านการสะกด — `EFX_CONTROL_RESILIENCE`)
    - `ITEM190_PILL_010` (โอสถไร้กลิ่นอายพรางลมหายใจ — `EFX_SCENT_TRACE_BREAK`)
    - `ITEM190_PILL_011` (โอสถบำรุงปราณอสูรคู่สัญญา — `EFX_BEAST_BOND_CHANNEL`)
    - `ITEM190_PILL_012` (โอสถต้องห้ามโลหิตเดือดแลกตบะ — `EFX_BLOOD_SACRIFICE_OVERDRIVE`, `lifespanCostYears: 4`)
    - `EFX_V234_TOXIN_TRANSCENDENCE` (พิษโอสถคลั่งพลัง — `toxinOverdrive`) & `EFX_V234_OVERDRIVE_SURGE`.
3. **Runtime Combat-Pill Pipeline**:
    - `ST.preFight` & `ECO.applyPill` override (`js/stats.js:657–681`), `CPE.pickPurchase` & `CPE.autoUse` (`js/cpeco.js:279–285, 419`), `COMBAT.startFight` (`js/combat.js`), and `VC.simulateGuardianEncounter` (`js/visitors_clues.js:1703`).

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2. Critical Defect 1: `[PILL-CB-01]` Eating Lunch Blocks All Pre-Fight Combat Pills (`stats.js:663`), While Friendly Spars & Remote Expeditions Loot `W.sect.stock` for Free (`stats.js:667`)!

2.1 Bug A — Any Active Non-Combat Buff (`e.g. professionMeal` or `mentorGuidance`) Blocks `ST.preFight` `100%` of the Time (`js/stats.js:663`)!

Look at line 663 of `js/stats.js`:

```javascript
ST.preFight = (W, n, hard) => {
  if (n.foe || !n.cult || (n.buffs && n.buffs.some(b => b.until > ST.now)))
return;
```

- **Catastrophic Interaction**:
    - Eating a canteen meal cooked by a Spirit Chef (`ECO.finishMeal`, `economy.js:405`) grants `{ id: 'professionMeal', kind: 'food', dur: 180 }` (3 hours). Receiving `mentorGuidance` (`cpeco.js`) lasts 2 hours. Fish dishes (`fishing.js`) last 3–4 hours.

    - Because line 663 checks **`n.buffs.some(b => b.until > ST.now)`** across *all* buffs rather than checking `b.kind === 'cbPill'`, **every disciple who ate a meal or received guidance in the last 3 hours is `100%` blocked from taking `P:CB_*` pills when raiders attack the sect or before fighting a Boss**!
- **Fix**: Check `n.buffs.some(b => b.until > ST.now && b.kind === 'cbPill')` (and stamp `kind: 'cbPill'` on combat-pill buffs)!

2.2 Bug B — Friendly Spars Burn War-Reserve Pills & Away Expeditions Teleport Pills Out of Home Stock (`stats.js:667–679`)!

Look at `stats.js:667 & 672`:

```
JavaScript
if (!keys.length && n.rank >= 1) Object.keys(W.sect.stock).forEach(k => { if
(isCb(k) && W.sect.stock[k] > 0) keys.push([k, 'sect']); });
...
if (id === 'BERSERK' && !(hard && (E(n, 'personality.aggression') > 55 || ...)))
return;
```

1. **Friendly Sparring Waste (`!hard` / `f.serious === false`)**:
    - When `hard === false` (friendly `spar`!), only `'BERSERK'` is skipped on line 672—meaning disciples **grab Tier-3 `P:CB_SPIRIT` / `P:CB_EYE` war-reserve pills out of `W.sect.stock` for `0 CP` and poison themselves (`+6 toxicity`) just for a friendly courtyard practice spar**!
    - **Fix**: At the very top of `ST.preFight`, if `!hard` (i.e., `!f.serious`, unless `f.kind === 'leadership'` or high-ambition `'tourney'` finals using *personal* `n.inv.pills` only!), **return immediately**!
2. **Telepathic Cross-Continent Stock Looting (`n.away === true`)**:
    - In `VC.simulateGuardianEncounter` (`visitors_clues.js:1703`), the party is **hundreds of miles outside the sect (`m.away === true`)**, yet `ST.preFight` checks `!keys.length && n.rank >= 1` and **teleports `P:CB_*` pills straight out of the home sect's `W.sect.stock` for `0 CP`**!
    - **Fix**: Require `!n.away` (and active Raid / Sect Defense emergency authorization: `W.threat && W.threat.raid`) before allowing free emergency draws from `W.sect.stock`; expedition members must pack `P:CB_*` in `n.inv.pills` beforehand!

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3. Critical Defect 2: `[PILL-CB-02] stats.js:658` Override Short-Circuits `ECO.applyPill` — Bypassing `daoOath` Vows, Pill Quality, and Multi-Effect Combat Pills!

Look at `js/stats.js:657–662`:

```javascript
const oApplyPill = ECO.applyPill;
ECO.applyPill = (W, npc, key) => {
  const info = ECO.pillInfo(key);
  if (!info || !info.effect.cbBuff) return oApplyPill(W, npc, key);
  const b = info.effect.cbBuff; ST.addBuff(npc, b);
  if (npc.cult) npc.cult.toxic = Math.min(100, (npc.cult.toxic || 0) +
(info.effect.toxicity || 0));
  HC.bump(npc); return { info, msgs: ['บัฟ「' + b.name + '」'] };
};
```

- **Four Compounded Bugs**:
  1. **Skips `VOW_ABSTAIN_PILLS` / `VEGETARIAN_PACIFIST` (`daoOath`)**: `ST.preFight` (`line 664`) only checks `voluntaryVows.VOW_NO_MEDICINE`, **forgetting `npc.daoOath`**, and line 659 returns **before** `oApplyPill` (`economy.js:339`) can check `npc.daoOath`! Thus an ascetic monk upholding `VEGETARIAN_PACIFIST` (while `P:CB_*` is literally brewed from `BEAST_PART` animal organs in `data_cb.js:18`!) swallows beast-blood combat pills with zero vow check!
  2. **Ignores Pill Quality (`q = 0.70` vs `q = 1.65`)**: `ST.addBuff(npc, b)` uses unscaled `b.p`, `b.f`, and `b.dur`—so a Masterpiece `q = 1.60` `P:CB_FURY` gives the exact same `+25% atk` as a burnt `q = 0.70` batch!
  3. **Drops Co-Effects on Hybrid Combat Pills**: Returning early on line 659 skips `healHealth`, `qiRestore`, `restoreBeastFatigue`, `cultivationProgress`, and `executeItemEffectById`!
  4. **Outdated Path Preferences (`line 669`)**: `SHADOW_SAMURAI` (`matk` useless!) falls into the default preference list containing `'SPIRIT'` (`+25% matk`), wasting war pills on magic attack! Derive preference dynamically from `(st.atk >= st.matk * 1.15)` $\to$ `['FURY', 'EYE', 'STONE', 'WIND']` vs `(st.matk > st.atk * 1.15)` $\to$ `['SPIRIT', 'WIND', 'STONE', 'EYE']`.

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4. Critical Defect 3: [PILL-CB-03] Zero Autonomous Purchase & Usage (0.00%) of Tactical, Stealth, Beast-Bond & Toxin-Overdrive Pills (ITEM190_PILL_009, 010, 011 & EFX_V234_TOXIN_TRANSCENDENCE)!

4.1 Exact Code Proof (js/economy.js:331–335 vs js/cpeco.js:265–286, 419 & js/stats.js:665)

Look at how ECO.pillInfo (economy.js:331–335) translates these 4 specialty pill families:

```javascript
else if (x.effectId === 'EFX_CONTROL_RESILIENCE') { e.grantBuff = e.grantBuff || 'clarityLotus'; e.buffDur = 75; }
else if (x.effectId === 'EFX_SCENT_TRACE_BREAK')  { e.grantBuff = e.grantBuff || 'shadowVeil'; e.buffDur = 60; }
else if (x.effectId === 'EFX_BEAST_BOND_CHANNEL') { e.grantBuff = e.grantBuff || 'beastPackFury'; e.buffDur = 75; e.restoreBeastFatigue = p.restoresBeastFatigue || 35; }
else if (x.effectId === 'EFX_V234_TOXIN_TRANSCENDENCE') { e.grantBuff = 'toxinOverdrive'; e.buffDur = 65; e.cultivationProgress = ...; }
```

Now check every AI consumer in the codebase:

1. **CPE.pickPurchase (js/cpeco.js:265–286)**: Checks healHealth, healInjury, clearToxic, clearHeartDemon, insight, btFor, **cbBuff**, and cultivationProgress. **It has ZERO check for e.grantBuff or e.restoreBeastFatigue!**
2. **ST.preFight (js/stats.js:665)**: Checks k.startsWith('P:CB_') || info.effect.cbBuff — **ignores e.grantBuff completely!**
3. **CPE.autoUse (js/cpeco.js:419)**: **Zero check** for restoreBeastFatigue when n.pets has a fatigued spirit beast (pt.fatigue >= 65)!

- **Result**:
  - ITEM190_PILL_009 (โอสถใจกระจ่างต้านการสะกด), ITEM190_PILL_010 (โอสถไร้กลิ่นอายพรางลมหายใจ), and ITEM190_PILL_011 (โอสถบำรุงปราณอสูรคู่สัญญา) have **0% autonomous NPC purchase and 0% autonomous consumption**!
- **Exact Emergent Wiring**:
  - **ITEM190_PILL_011 (restoreBeastFatigue)**: In CPE.pickPurchase & CPE.autoUse, BEAST_SOVEREIGN disciples (or any pet owner with n.pets.some(p => p && !p.dead && p.fatigue >= 55)) actively buy and administer restoreBeastFatigue pills!

- **ITEM190_PILL_010 (EFX_SCENT_TRACE_BREAK / shadowVeil)**: Scouts (`post === 'scout'`), `NINPO_SHADOW`, `WANDERING_DAO`, and Clue/Secret-Realm expedition leaders buy and consume it before departing (`-25%` ambush rate in `explore.js` / `expedition.js` + `shadowVeil` in Round 1)!
- **ITEM190_PILL_009 (EFX_CONTROL_RESILIENCE / clarityLotus) & EFX_V234_TOXIN_TRANSCENDENCE (toxinOverdrive)**: Include combat-oriented `grantBuff` pills in `ST.preFight` (`when hard === true`), with `VENOM_GU` / `DEMON` (`c.toxic >= 30`) strongly favoring `EFX_V234_TOXIN_TRANSCENDENCE`!

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5. Critical Defect 4: [PILL-CB-04] Forbidden Blood-Sacrifice Pill (ITEM190_PILL_012 โอสถต้องห้ามโลหิตเดือดแลกตบะ) Deducts 0 Lifespan & Is Eaten Casually by Orthodox Monks!

5.1 Exact Code Proof (data/data_items.js, js/economy.js:334, js/cpeco.js:285)

- `ITEM190_PILL_012` declares: `"effects": [{ "effectId": "EFX_BLOOD_SACRIFICE_OVERDRIVE", "params": { "powerMultiplier": 2.35, "healthCost": 18, "lifespanCostYears": 4 } }]`, `"descTh": "โอสถต้องห้ามที่เผาแก่นโลหิตและอายุขัยสี่ปีเพื่อเร่งพลังรบและตบะชั่วพริบตา"`
- **Live Bugs**:
  1. **Zero Lifespan Cost & Zero HP Cost (economy.js:334 & 857)**: Consuming `ITEM190_PILL_012` burns **0 years of lifespan** and **0 HP**!
  2. **Eaten as Casual Peace-Time Exp Candy by Righteous Monks (cpeco.js:285)**: Because `economy.js:334` adds `e.cultivationProgress = 28 * it.tier` (`+140` exp) with **no taboo: true flag**, `CPE.pickPurchase` (`line 285`) treats it as an ordinary cultivation pill—so a peaceful `BUDDHA_DHARMA` monk under `STRICT_ORTHODOX_BAN` casually buys and swallows a Forbidden Lifespan-Burning Blood Pill on a sunny afternoon!
- **Exact Fix**:
  1. Mark `e.taboo = true; e.lifespanCostYears = p.lifespanCostYears || 4; e.hpCostPct = (p.healthCost || 18) / 100;` in `ECO.pillInfo`.
  2. Deduct `lifespanCostYears` (`npc.ext.pillLifespanYears -= 4` / `c.lifeBonus -= 4; CULT.refreshLifespan(npc)`), `-18%` current HP, `+8`

`tabooStrain`, and `-10 sinKarma` in `ECO.applyPill` & `executeItemEffectById`!

3. Block `e.taboo` purchases/consumption in `CPE.pickPurchase` & `ST.preFight` when `W.sect.forbiddenVault?.policy === 'STRICT_ORTHODOX_BAN'` (unless low-loyalty rebel / `DEMON` / `VOID_HERETIC_PACT`, or desperate `serious: true` life-or-death crisis at `HP < 22%`)!

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6. Mandatory Verification Suite Additions (`06D`)

2. **Lunch-Buff Non-Interference Test**: Give an NPC `professionMeal` (`kind: 'food'`) and a `P:CB_FURY` pill, then run `ST.preFight(W, npc, true)`; assert `P:CB_FURY` **is** consumed and grants its combat buff alongside `professionMeal`.
3. **Spar & Away-Stock Isolation Test**: Assert `ST.preFight(W, npc, false)` consumes `0` pills from `W.sect.stock`, and an `npc.away === true` expedition member consumes `0` pills from `W.sect.stock`.
4. **Vow & Taboo Lifespan Test**: Assert `CULT.isAbstainingFromPills(npc)` blocks `ST.preFight`, and consuming `ITEM190_PILL_012` reduces `lifespanMax` by `4` years and respects `forbiddenVault.policy`.
5. **Beast-Bond & Tactical Pill Autonomy Test**: Give a `BEAST_SOVEREIGN` NPC a fatigued pet (`fatigue = 75`) and `ITEM190_PILL_011`; assert `CPE.autoUse` consumes `ITEM190_PILL_011` and reduces `pet.fatigue` by `35`.